The empirical recurrence for OEIS A301845, recovered from its tail

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Abstract

OEIS A301845 records "Empirical recurrence of order 99" and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 300$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 6$, so the recurrence is proved for every $n \geq 105$. Its last coefficient is $c_{99} = 8 \neq 0$, so no further tail satisfies anything shorter; any order-99 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry's recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A301845 is "Number of 5Xn 0..1 arrays with every element equal to 0, 1 or 2 horizontally or antidiagonally adjacent elements, with upper left element zero.". It has offset 1 and begins

$$16, 512, 4299, 48802, 496085, 5240684, 54948498, 577194574, \ \ldots$$

The entry states:

> Empirical recurrence of order 99 (see link above)

As of the "Last modified" line on the live entry (Mar 27 2018 14:37:28, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry's condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u, and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) \;=\; \iota^{\top} M^{\,j(n)} \tau$$

for a linear reindexing j fixed by the entry's shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 300$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 300$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \ldots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S'+1$ consecutive values of a. $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 99: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 6$ is the first whose minimal order is exactly the stated 99.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 600$ exact terms from $n = 6$ on, it returns order 99 — the order the entry states — with integer coefficients:

$$a(n) \;=\; \sum_{i=1}^{99} c_i\, a(n-i),$$

c_{1}	17	c_{26}	30036726611	c_{51}	−709940968844	c_{76}	239455095
c_{2}	−72	c_{27}	13676489983	c_{52}	932528303264	c_{77}	1573424745
c_{3}	3	c_{28}	121964398820	c_{53}	−484755172330	c_{78}	779750014
c_{4}	418	c_{29}	−308012302397	c_{54}	−164085932592	c_{79}	472701109
c_{5}	−261	c_{30}	−52705416714	c_{55}	−40610621267	c_{80}	266808236
c_{6}	1072	c_{31}	898722538874	c_{56}	6028677744	c_{81}	31437711
c_{7}	−10546	c_{32}	−1369298839640	c_{57}	27308585295	c_{82}	−25987223
c_{8}	4531	c_{33}	1689297453239	c_{58}	540036609973	c_{83}	−51599614
c_{9}	88464	c_{34}	−2637409509064	c_{59}	112922083450	c_{84}	−42043839
c_{10}	−135279	c_{35}	2149388358944	c_{60}	480833078801	c_{85}	−2143061
c_{11}	−464537	c_{36}	−125732489258	c_{61}	384339741008	c_{86}	4685328
c_{12}	928039	c_{37}	−673163556117	c_{62}	−625862102329	c_{87}	970466
c_{13}	2880566	c_{38}	1003116414681	c_{63}	−241654754467	c_{88}	1396144
c_{14}	−4484894	c_{39}	−2358715190684	c_{64}	90180677811	c_{89}	648883
c_{15}	−12630498	c_{40}	3466861196998	c_{65}	−338414819932	c_{90}	262331
c_{16}	34924472	c_{41}	−2814485763548	c_{66}	−107679737917	c_{91}	−32699
c_{17}	−94176144	c_{42}	2534964856459	c_{67}	171461383246	c_{92}	−77472
c_{18}	−90846940	c_{43}	−3263102466872	c_{68}	69088126682	c_{93}	−24983
c_{19}	1471971377	c_{44}	4219099011331	c_{69}	69280244087	c_{94}	−5530
c_{20}	−1321195730	c_{45}	−3494075472928	c_{70}	40893644486	c_{95}	−606
c_{21}	−6868259637	c_{46}	2592735332772	c_{71}	−11047478349	c_{96}	306
c_{22}	9871666327	c_{47}	−1455957725252	c_{72}	−4195582024	c_{97}	252
c_{23}	16944079844	c_{48}	−595196571071	c_{73}	−10971996474	c_{98}	80
c_{24}	−28670082557	c_{49}	203136733833	c_{74}	−13188427971	c_{99}	8
c_{25}	−22948802844	c_{50}	18106783544	c_{75}	−5107698476		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 105$, and it is the recurrence OEIS A301845 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 6$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{99} - c_1 x^{99-1} - \cdots - c_{99}$. Its constant term is $-c_{99} = -8$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 99 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 99, so $\deg q = 99$, and it is monic. It holds from some index t on. From $\max(t, 6)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 99, so $q = q_{\min}$. □

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 17 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry's English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 99, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry's.

Third, the recovered recurrence was evaluated directly on the entry's own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 8.

Remark 4. The conjecture's own statement was never read. The entry pins it down to the family of order-99 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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